

Introduction

Our research addresses a **combinatorics** problem on identifying right triangles on a $\mathbb{Z}_n^2$ grid. We investigate the minimum number of points required to guarantee the existence of a right triangle regardless of configuration, and we construct an efficient algorithm to identify a right triangle under point updates. This research relates to classic combinatorics problems such as the No-Three-in-Line Problem [2] and the Corners Theorem [1].

Proof by Contradiction

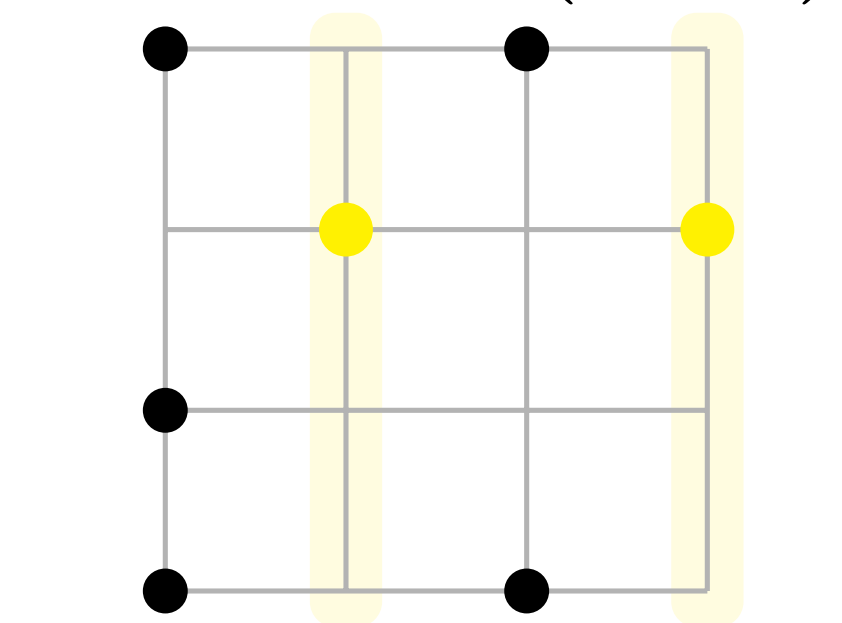
Preliminaries:

- A point P is in row i and column j .
- A point is a corner iff $|row_i| > 1$ and $|col_j| > 1$.
- A row/column is *light* if there are no corners in it.

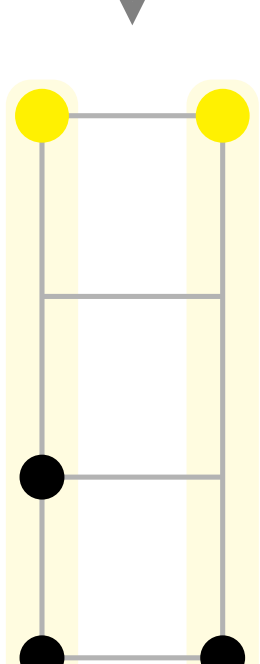
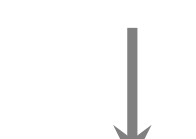
Hypothesis: Assume that there is no right triangle one can make given $2n - 1$ points in a $\mathbb{Z}_n^2$ grid. Then all $2n - 1$ points must be placed in a light row and column.

- Pigeonhole Principle:** There is guaranteed to be at least one row with $\lceil \frac{2n-1}{n} \rceil = 2$ points.
- Column Reduction:** Because all points are placed within a light row/column, these points cannot be corners. Therefore, neither of these points can have other points in their respective columns. Remove these two points and columns.
- Contradiction:** Repeatedly iterating through this process will eventually leave us with either $n - 1$ points and 0 columns (if n is even) or n points in 1 column (if n is odd). The first case is impossible, and the second case contradicts the hypothesis since this entire column must then be filled up, which guarantees the existence of a corner.

Even Case ($n = 4$)



Step 1: 4 columns, 7 points.

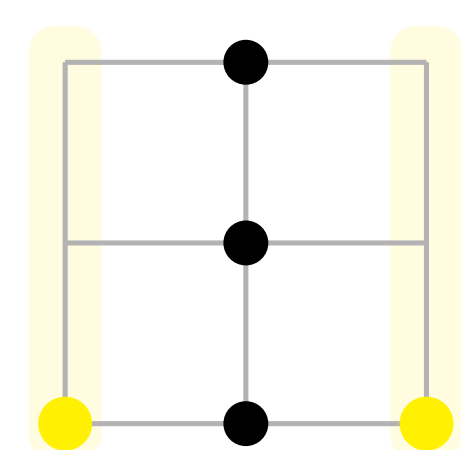


Step 2: 2 columns, 5 points.

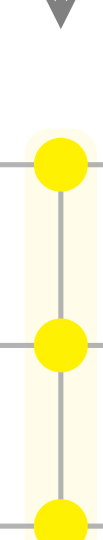


Contradiction: 0 columns, 3 points.

Odd Case ($n = 3$)



Step 1: 3 columns, 5 points.



Contradiction: 1 column left, so all 3 points must share it. At least one of these rows must contain another point.

Algorithm

Data Structures:

- A **list** of all corners
- A **list of lists** row where row_i contains all corresponding points in row i .
- A **list of lists** col where col_j contains all corresponding points in column j .
- A **hash table** of all points. A coordinate (i, j) maps to a 3-tuple (c, x, y) where:
 - c : the index in the corners list (-1 if it isn't a corner)
 - x : the index in list row_i
 - y : the index in list col_j

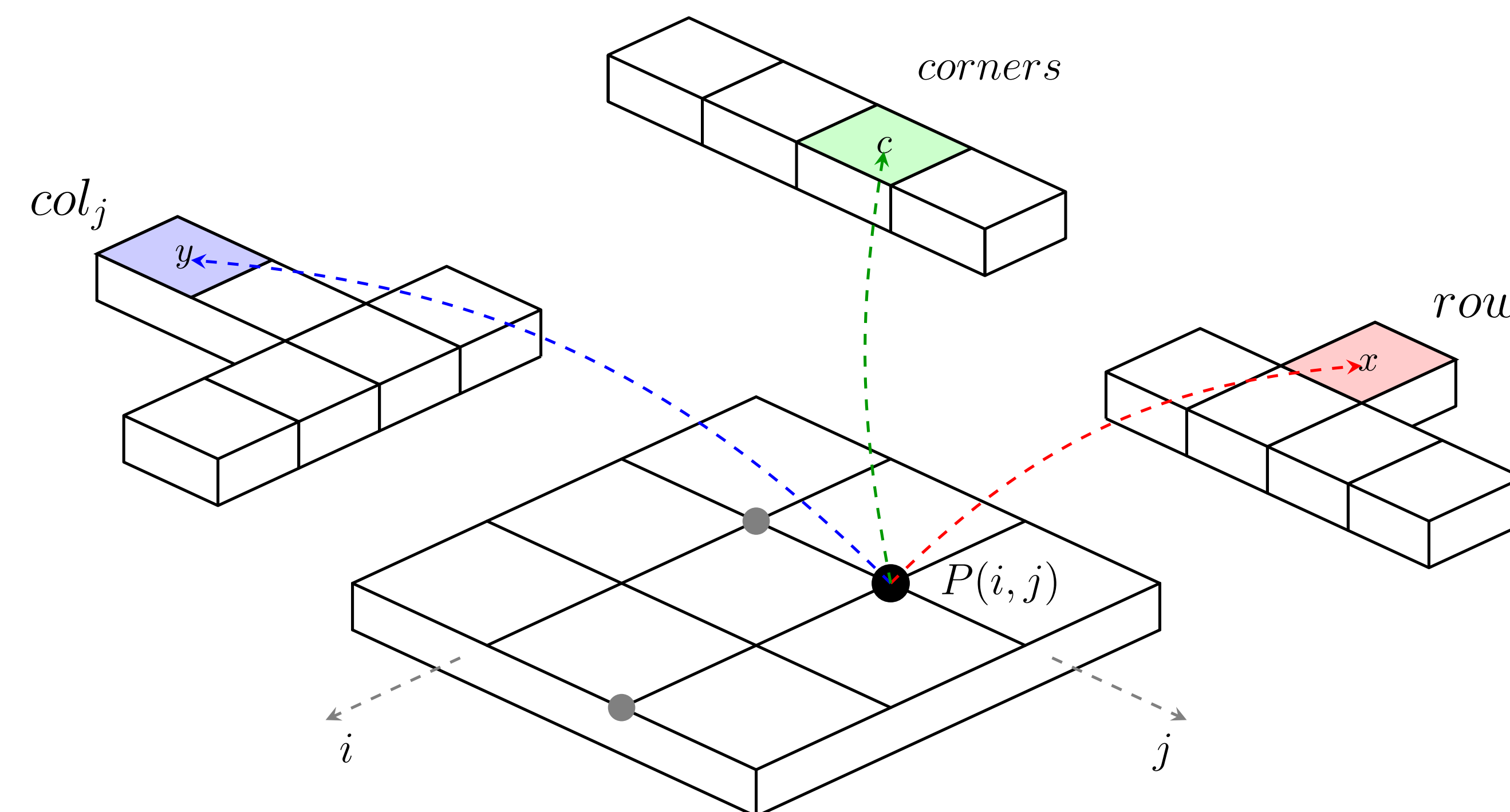


Fig. 2: Point with respect to data structures

Operations:

- Moving a Point:**
 - Add a Point:** Add the point to the hash table and row/column lists. Confirm if the point is a triangle; if so, add to corners list.
 - Delete a Point:** Delete the point from the row/column lists and corners list if applicable. Remove from the hash table.
 - Verify Adjacent Points:** When adding a point, if the size of either the row/column list becomes 2, verify if any adjacent point also becomes a corner. Conversely, when deleting a point, if the size of either list becomes 1, verify if any adjacent point is no longer a corner. Update data structures accordingly.
- Retrieving a Triangle:** Take a point from the front of the corners list. Identify a unique point from its respective row and column lists (≤ 2 operations each).

Time and Space Complexities

Time Complexity: Expected $O(1)$

- Finding a Triangle:** Retrieve a point from the corner list and retrieve 2 other points from its row and column lists. List indexing is $O(1)$.
- Moving a Point:** Hash table operations (lookup/delete) and list insertions are $O(1)$ (more specifically, amortized $O(1)$ for list resizing and expected $O(1)$ given a good hash function). List deletions are also $O(1)$ using swap and pop.

Space Complexity: $O(N)$

- We only store information regarding the points and not grid cells. This scales linearly with N .

Visualization

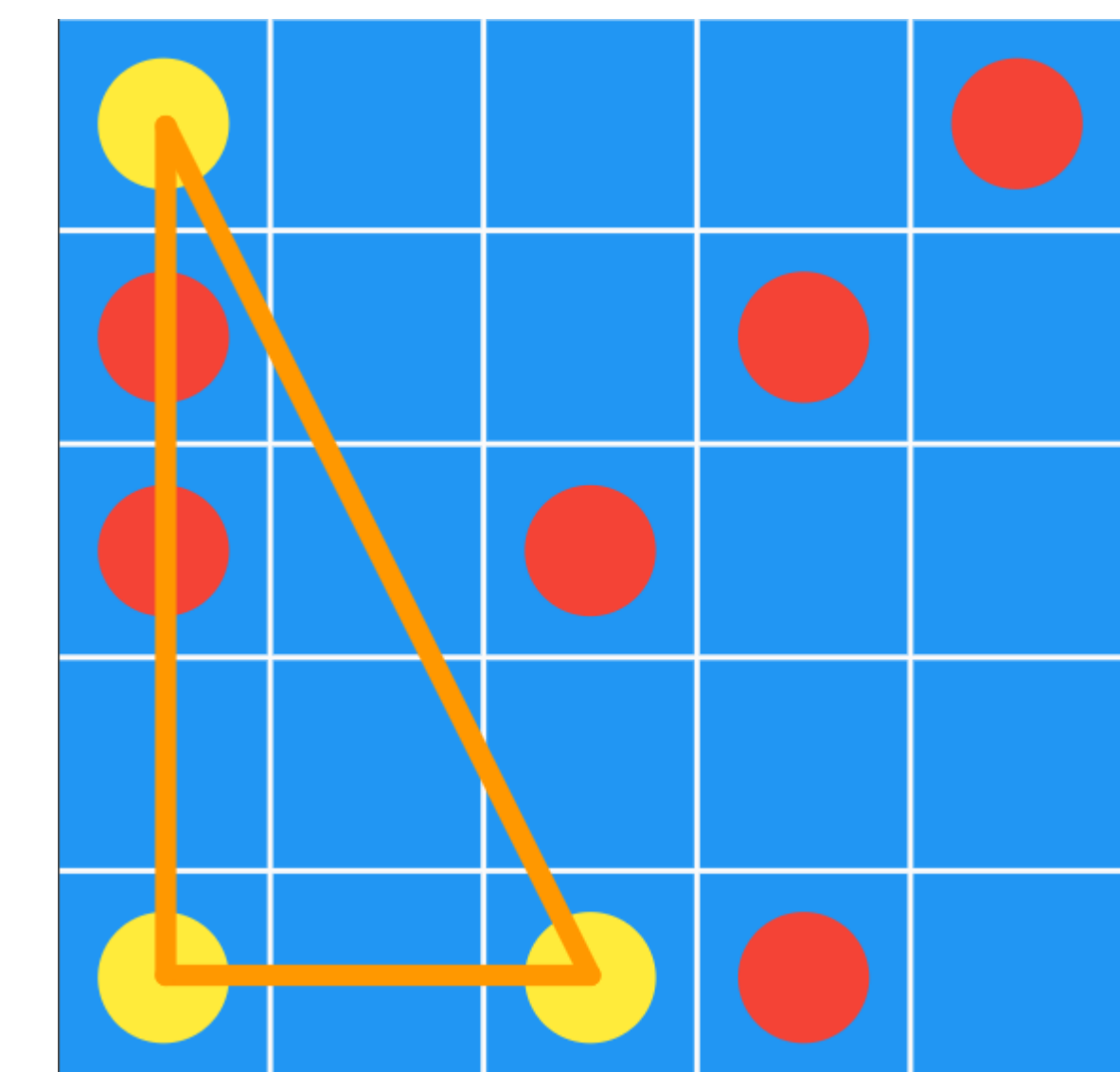


Fig. 3: Interactive visualization on Saad's website [3]

Future Work

There are many research problems one can explore as a continuation to this setting. Some are:

- Identifying the minimum number of triangles in any given grid
- Generalizing to a $\mathbb{Z}_m \times \mathbb{Z}_n$ grid
- Developing an algorithm to efficiently enumerate all possible configurations of a grid

References

- [1] M. Ajtai and E. Szemerédi. "Sets of lattice points that form no squares". In: *Studia Scientiarum Mathematicarum Hungarica* 9 (1974), pp. 9–11.
- [2] R. R. Hall et al. "Some Advances in the No-Three-in-Line Problem". In: *Journal of Combinatorial Theory, Series A* (1975).
- [3] S. Mneimneh and J. Lee. "Right Triangles on a Grid Interactive Visualization Tool". In: (). URL: <https://www.cs.hunter.cuny.edu/~saad/triangles/web/>.